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Understanding vision-based continuous sign language recognition

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Abstract

Real-time sign language translation systems, that convert continuous sign sequences to text/speech, will facilitate communication between the deaf-mute community and the normal hearing majority. A translation system could be vision-based or sensor-based, depending on the type of input it receives. To date, most of the commercial systems for this purpose are sensor-based, which are expensive and not user-friendly. Vision-based sign translation systems are the need of the hour but should overcome many challenges to build a full-fledged working system. Preliminary investigations in this work have revealed that the traditional approaches to continuous sign language recognition (CSLR) using HMM, CRF and DTW, tried to solve the problem of Isolated Sign Language Recognition (ISLR) and extended the solution to CSLR, leading to reduced performance. The main challenge of identifying Movement Epenthesis (ME) segments in continuous utterances, were handled explicitly with these traditional methods. With the advent of technologies like Deep Learning, more feasible solutions for vision-based CSLR are emerging, which has led to an increase in the research on vision-based approaches. In this paper, a detailed review of all the works in vision-based CSLR is presented, based on the methods they have followed. The challenges posed in continuous sign recognition are also discussed in detail, followed by a brief on sensor-based systems and benchmark databases. Finally, performance evaluation of all the associated methods are performed, which leads to a short discussion on the overall study and concludes by pointing out future research directions in the field.

Keywords Continuous sign language · Vision-based · Movement epenthesis · Review

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1 Introduction

Sign language is the movement language that the hearing and speech impaired community uses for expression of their thoughts and emotions through a series of hand and arm motions (manual signs), facial expressions and head/body postures (non-manual signs). Sign language is regarded as structured, constrained by the rules of syntax [43, 66]. Sign languages are natural languages, with lexicon and grammar and therefore not a world-wide language [80]. Popular sign languages are British Sign Language (BSL) in the UK [85], American Sign Language (ASL) in the US [61], Indian Sign Language (ISL) in India [89], AUSLAN in Australia [42], etc. Thus, different countries have developed, or are in the process of developing their sign language dictionaries. Signs are performed either with one or both hands. A limited set of people with normal speech faculty understand sign languages. To facilitate smooth communication between the hearing-impaired or deaf and the majority, outfitted with normal hearing faculty, automatic Sign Language Recognition (SLR) systems are the need of the hour. Recognition of sign language requires simultaneous interpretation of multi-modal data - facial expression, body posture, hand pose and hand movement [7, 103]. Real-time translation, of sign language in a specific SLR system to text or speech, can be an intricate task. Extensive studies in SLR systems have shown notable commonality with the domains of machine analysis, and interpretation of human actions and behavioral expressions [27, 50, 74, 74, 97].

SLR can be further classified into three sub-problems:

1. static sign language recognition (SSLR) - recognition of fingerspelled gestures of alphabets and digits [53, 71, 87, 101].
2. isolated sign language recognition (ISLR) - recognition of dynamic gestures corresponding to words [59, 62, 64, 106].
3. continuous sign language recognition (CSLR) - recognition of sequences of dynamic gestures, corresponding to sentences interspersed with Movement Epenthesis (ME) (non-sign sequence between end of a sign and start of another sign) segments as well [11, 33, 37, 47, 100, 104].

An ideal CSLR system should produce a transcription of the sign utterance, either as speech or text. Based on the processing technique, the system takes input in the form of video (vision-based system) or sensor data (glove/sensor-based system). This is depicted pictorially in Fig. 1. While the tasks of SSLR and ISLR are to identify the precise sign label,

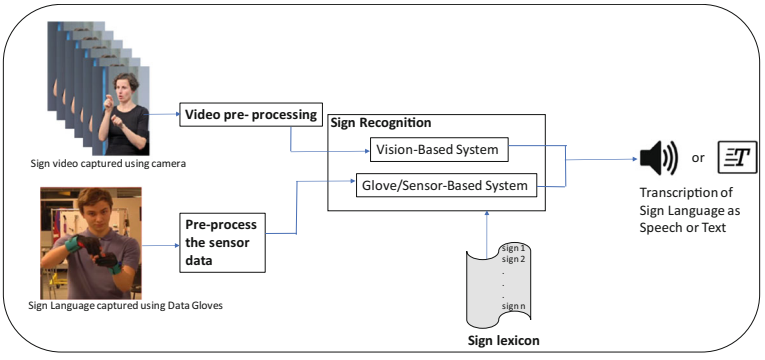


Fig. 1 Overview of a Sign Recognition System

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continuous SLR involves the omission of MEs and the segmentation of dynamic signs, corresponding to words, followed by the assignment of the correct label. SSLR and ISLR are dealt as independent problems in literature. But it also comes as a sub-problem, when dealing with the larger problem of CSLR. This is because, continuous utterances consist of gestures corresponding to words as well as alphabets/digits (fingerpelling). Constrained by the effects of ME and various other challenges like co-articulation (detailed below), the solution to ISLR and SSLR often does not generalize or extend to CSLR. To handle these critical problems of CSLR, statistical and machine learning methods have been proposed and implemented. Researches in this direction should be critically analyzed for identifying shortcomings and provide guidelines for future work. This paper is mainly focused on vision-based approaches of CSLR. Related works are reviewed in detail, where the taxonomy is based on the schemes adopted by authors of different papers over time. Albeit no survey has been done on sign languages with this aforesaid intent, few related studies are covered in [1, 72, 116].

The rest of this paper is organized as follows: Section 2 describes the challenges posed in the continuous sign recognition scheme. Section 3 presents an overall idea of a sign recognition system, wherein Section 3.1 briefly reviews glove-based recognition systems; Section 3.2 covers a detailed survey of vision-based recognition systems. Section 4 dwells on the various benchmark databases for CSLR, available in different languages. Section 5 reports on performance evaluation, followed by Section 6, which discusses the findings related to CSLR. Section 7 has concluding remarks, summarizing the state-of-the-art and future research opportunities in this field.

2 Challenges in continuous sign language recognition

CSLR is directed to development of good models for accurate identification and labeling of a sequence of signs. Majority of the prior studies have dealt with getting a handle on the task of ISLR. Nevertheless, continuous sign recognition scheme presents more challenges than isolated sign recognition, resulting in a low recognition rate for real-world data. The challenges faced in CSLR are detailed below.

(i) *Feature extraction from a complex background*: The task of feature extractions, from complex backgrounds, is very crucial for a real-time sign recognition system, since the final outcome depends on these hand-engineered features. A greater number of conventional investigations create synthetic datasets, with uniform backgrounds, to simplify the process of hand segmentation. Few others use colored gloves for identification of the prominent hand. The challenges encountered in the realization of good segmented features have prompted many researchers to use non-vision based alternatives such as Data gloves, with built-in sensors. Vision-based approaches, either use skin color or track motions, to detect hands. Color-based hand segmentation approaches are more sensitive to illumination. Motion tracking techniques are based on the presumption that the relevant features for SLR are associated with motions of the hand. Shape modeling [84] and particle-filtering [81] are two ways of tracking hands, in a noisy environment; but, these methods require initialization and hands can still be lost due to occlusion. Better segmentation results can be achieved with fusion [4] or multi-modal [31] approaches. [56, 109, 111] are some of the early CSLR probes, reported in the literature, which employed feature engineering for recognition purposes. Also, the sign classification algorithm should be capable of

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differentiating between hand shapes, even in the case of large lexicons, and provide good recognition of shapes ; development of such an algorithm poses a formidable exercise. Thus, the task of feature extractions pose a big challenge to error-free sign recognition.

(ii) *Large vocabulary and Scalability*: Another daunting challenge in sign language recognition pertains to definition of a modeling paradigm, powerful enough to capture the language, yet scalable to large vocabularies. Few works that dealt with this challenge are [21, 44]. Just like a Natural Language Processing (NLP) system with large lexicon, an ideal sign recognition system should offer vocabulary of similar size. But as the size of vocabulary increases, the inter-class variations are reduced naturally, which may lead to misclassifications in the recognition phase, culminating in lower detection rate for the final CSLR system. An ideal scalable recognition system is the one that shows consistent performance even when the lexicon is expanded. Most of the systems reported in the literature, have employed sign-level, context-dependent Hidden Markov Models (HMMs) or Conditional Random Fields (CRFs), based on maximum likelihood estimation, thereby resulting in an increased computational complexity on increasing the size of the vocabulary. This in turn affects the scalability of systems.

(iii) *Movement epenthesis (ME)*: Transition segments, called ME, are formed in sign sequences, which connects successive signs when the hands move from the ending location of one sign to the starting location of the next sign [13]. ME does not carry any information on a sign. No signs are associated with these ME frames, which could even span in length equal to actual signs. Published literature do not afford a well-defined lexicon of MEs, and related linguistic studies are limited. Given N possible signs in a vocabulary, there would be $O(N^2)$ possible types of ME, which make it computationally burdensome to model explicitly, all possible MEs ([44, 93, 109, 110, 112]); the situation would be much worse with increase in the size of a vocabulary. Most of works in CSLR dealt with this problem [21, 26, 44, 56, 93]. Since ME is a connecting segment between signs, its starting and ending locations would depend on the preceding and succeeding signs; variations in adjacent signs may affect the ME and vice-versa. Dealing with ME adds significant complexity to the task of recognizing continuous signs, as there are no well-defined rules for making such transitional movements. This could be considered as the fundamental problem that needs to be addressed explicitly, for robust SLR. In short, SLR requires segmentation of MEs, either implicitly or explicitly, which renders CSLR an arduous task.

(iv) *Signer-Dependent variations*: Signer-specific variations are variations arising from varying signers, who gesticulate naturally. These variants could be either grammatical or phonological (i.e. variations in hand shape, movement, palm orientation and location). Intra-/inter-personal variability may lead to different signing speeds. The signing speed should be moderate for proper recognition of sign linguistics. Slow signing increases the frames per second, while high signing speed emanate as information loss, motion blur, deletion error, and the like. Therefore speed synchronization presents an onerous challenge for CSLR systems. Two strategies that could be employed for signer-independence, in recognition systems, are: (1) a baseline recognition system could be built with a few signers, and by usage of a small amount of training data, the system could be adapted to a new signer, or (2) a robust recognition algorithm, tolerant to signer variations, could be devised for good

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generalization. Few works in CSLR have actually dealt with the problem of signer-induced variations [44, 51, 95]. However, the task, dealing with real-world data, continues to be a hard nut to crack.

(v) *Sub unit modeling*: Sign-level recognition becomes intractable as the size of vocabulary increases. A potential alternative is to identify the sub-components of signs, and combine them to form sign words. Thus, a sub-unit based SLR system, adopts a two-stage recognition process - (Step 1) identification of sign linguistic sub-units, (Step 2) creation of a sign-level classifier, by combining the sub-units. Sub-unit modeling could not only efficiently handle large lexicons, but also be tolerant to sign variations, due to intra- and inter-signers. Few SLR systems that have adopted sub-unit modeling are noted in [11, 14, 35, 36, 47, 77, 79].

(vi) *Occlusion*: For automatic SLR, three objects need to be tracked: signer's face and two hands. The task of tracking is very difficult since face and hands are of the same skin color and often overlap while signing (hand-face occlusion), from the camera view-point. In such cases, hand segmentation based on skin color is impractical, since both hand and face have similar skin tones. [82, 105] and [30] tried to tackle this problem and were successful to some extent. Also, in many cases, the configuration and identification of both hands are required for correctly identifying a sign. When the hands are occluded or touch each other (hand occlusion) and in case both hand shapes are required, it results in a single connected-component that leads to difficulty in segmentation. Due to high chances of occlusion and cross-over, the left hand might be mistakenly identified as right, and vice-versa. For dealing with this type of occlusion Random Occupancy Pattern (ROP) [98] and 3D/4D space of time (HON4D) [73] were proposed. Researchers dealing with dynamic sign recognition should deal with the occlusion problem in the feature extraction stage itself.

(vii) *Integrate local and global features*: Continuous sign language is a case of dynamic gesturing having both local and global motions. For improved recognition, it is better to combine global hand motion and local motion [28]. Global hand motion corresponds to motion of centroid of the gesture and local motion is the motion of the fingers w.r.t the centroid. Integration of both the features are essential for CSLR but it is a difficult task.

(viii) *Co-articulation effect*: Analogous to "co-articulation" issue in speech [24], co-articulation occurs in sign language as well, where the appearance of a sign is affected by the previous and successive signs. Each unit of a fingerspelled sign or continuous sentence is affected by the other units, preceding and succeeding it. The unit of a sentence or word is rotated, pulled, stretched, or twisted by the previous and the following units. "Co-articulation"-like processes occur only in a small number of signs, unlike ME, which occurs very frequently between consecutive signs [72]. Bhuyan et al. [6] efficiently utilized motion features to address the problem of co-articulation. The authors of this paper concur with Sylvie and Ranganath's position [72], that ME should be dealt with first. However, due to co-articulation, there are high chances of complete omission of hand shapes, leading to misinterpretation of signs.

(ix) *Sub sign reasoning*: Different signs may share some patterns at times, and signs that appear as part of other signs (super signs) are called sub signs. Yang et al. [109, 110] have tried to address the problem of sub-signs in the context of continuous sign recognition. On encountering such signs, a CSLR system should correctly identify the super sign, which can pose an additional challenge.

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(x) *Short sign detection and Sign length constraint*: Signs of short duration like that of fingerspelled signs are called short signs. Even the dynamic signs, if signed in speed, may result in short signs. Yang et al. [109, 110] have tried to address this problem in the context of continuous sign recognition. It is difficult to recognize short-signs compared to long signs since the former has fewer frames. During sign recognition, there is a high chance of missing some of the frames corresponding to short signs. Missed frames in turn, pave the way for deletion errors, complicating the problem of accurate sign recognition. For the employment of Viterbi-based HMM/CRF [18], all signs should be of uniform duration, for which, a constraint on the sign length is imposed. The following are a few latest works [11, 16, 48] that constrained sign length. But with Deep Learning methods, the sign length is not constrained, instead, frame-alignment is done to map the labels corresponding to each word (set of frames) in the gesture sequence.

(xi) *Modality combination*: Sign languages convey meaning through multiple channels or modalities at the same time, such as body pose, facial expression and hand motion. Both manual and non-manual features need to be studied for complete recognition of a continuous sign. Hand shape and orientation contribute to manual features whereas facial expressions, head/body postures contribute to non-manual features. With this approach, many ”phonemes” are produced simultaneously, making sign languages highly non-sequential. The non-sequential aspect of sign languages poses a challenge to statistical sign language recognition systems. Few works that experimented on modality combinations are [17, 47] and [104]. The challenge involved with modality combination is that the feature extraction technique should combine data from all the channels in an efficient way.

(xii) *Fingerspelled signs interleave dynamic signs*: Hand shape, motion and location w.r.t face, and body of the signer, are important for CSLR. Fingerspelling spotting is based on hand-shapes and hand location only, whereas continuous sign recognition is based on all the three features, i.e. hand shape, motion and location w.r.t face, and body of the signer. The internal structure of fingerspelling and signs are different; therefore, it is a challenge to simultaneously identify both. Yang et al. [110] developed an approach for simultaneously spotting dynamic signs and fingerspellings. Moreover, fingerspellings are static and very short duration signs and when it appears between dynamic signs, it is even more challenging to identify them as fingerspellings.

3 Literature review: sign recognition systems

A sign recognition system is capable of accepting input signs, either in the form of video or as sensor data. The input signs are then processed and translated into text or speech. If the input data (continuous signing) is captured as video, then it is a vision-based recognition system; else, if the system works on sensor data, then it is a glove-based or sensor-based recognition system. Further processing such as feature extraction and sign recognition are done on these video frames in case of a vision-based system. For a glove-based system, the sensor data corresponds to the bend angles for fingers, the abduction between fingers, the orientation (roll, pitch, and yaw) of the wrist etc., that not only ease the feature extraction process, but also results in accurate feature representations leading to better recognition rates.

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Earlier (before 2017), most of the research in CSLR were glove-based, since accurate features could be obtained from sensors. Vision-based recognition systems face many difficulties for efficient feature extraction and sentence segmentation. Even the representation of frames in the frequency domain fails to provide details of local aspects like hand position, shape, orientation and movement. The location of spatially relevant regions of an image is an inescapable, tedious task, required for the construction of appropriate features. For instance, it is hard to segment or track hands, even in a simple background, due to mutual occlusion of hands and the intricacy of localizing a hand dis-positioned in front of a face. These complex segmentation and feature extraction issues prompted SLR systems to lean on assistive tools, with sensors to obtain clean features. Achievement of the latter led to better recognition results as well. Although usage of data gloves yield better results, the signer is burdened with such assistive devices. Besides, this method is relatively expensive. Commercialization of a SLR systems call for adoption of cost-effective, in addition to being user-friendly. Vision-based recognition systems meet the required criteria, albeit at the cost of inaccurate features ensuing in compromised results. Such systems capture data as images or videos, using one or more of webcam, standard video camera, thermal camera, stereo camera, or more advanced active techniques such as Kinect and LMC. Stereo cameras, Kinect and LMC are 3D cameras, which can collect depth information.

Based on these learning, a brief review of glove-based systems is presented, followed by a detailed study of vision-based systems, which is the focus of this paper.

3.1 Glove-based recognition systems

In these systems, the data required for decision-making are captured through data gloves, outfitted with a variety of sensors: flexion sensors, accelerometers (ACCs), proximity sensors, and abduction sensors. A review of the advantages of glove-based system over vision-based system was detailed by Mohamed Aktham Ahmed et al. [1].

Using data gloves, Liang et al. [58] collected data for position, posture, orientation and motion. Based on the captured data, position model, posture model, orientation model and motion models were designed using HMMs and trained for CSLR by integrating them, on a vocabulary of 250 words. The discontinuities in input sentences were detected, by time-varying parameter detection, for segmentation of individual gestures. They achieved an average recognition rate of 80.4%. However, the system can identify word boundaries only if the signer signs at a slow pace. Ma et al. [26] deployed data gloves as well, to capture typical positions, orientations and postures for each hand as sub-words which are used as the smallest recognition units and states of HMM. For handling ME, Ma's team adopted sentence level training, where Dynamic Programming (DP) was employed to segment out sub-words, followed by combination of ANN(Artificial Neural Network) and DP to optimally code the sign. This context-dependent model, constrained by bigram language model, adopted a fast matching strategy for the decoding process, which resulted in a recognition rate of 95.2%. Subsequently, they proposed a context-independent recognition system by modification of HMMs for modeling sub-words and constraining viterbi into a single model; Ma's team achieved an accuracy of 70% when tested with 40 continuous signs [113].

Similar to Volger et al. [93], Gao's team [21] explicitly modeled ME. They proposed a temporal clustering to cluster ME frames, concomitant with the training data distribution. Based on the clustered data, transition movement models (TMMs) were trained, and sign models with segmented sign data, and isolated sign data were used for segmentation

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of continuous sentences into signs and MEs, which yielded a recognition rate of 91.9%. Kong et al. [51], using trackers and data gloves, captured raw data to adopt a multichannel, segment and merge, approach for CSLR, with the specific goal of signer-independence. Continuous sentences were first segmented, and the sub-segments were assigned SIGN/ME labels, by CRF and SVM. Their output probabilities were fused to a Bayesian Network to improve the accuracy. Next, the ME sub-segments were discarded, and the SIGN segments were merged and recognized, by a two-layer CRF model. When the model was trained with 8 signers, they obtained a recall rate of 86.6% and precision of 89.9% for unseen signers.

Wang et al. [100] modeled a sign as a 3 layer Hierarchical Grassmann Covariance Matrix (HGCM) on the Grassmann manifold [99]. A SVM classifier was then trained for sign matching, based on the labels assigned in HGCM model. The classifier used the Grassmann metric [34] to measure the distance between two sign language sequences, since the metric is capable of preserving the topological structure and also the representations on the Grassmann manifold is more compact. A Multi-Temporal Belief Propagation (MTBP) spatial graph was designed for segmenting and spotting the signs from continuous gestures. They reported a WER of 0.449 (log_2) on the dataset collected using Kinect sensor, as part of their work.

Since our focus is on vision-based systems, we limit the review of glove-based systems. From a brief study of these methods it is clear that, such systems fail to study the hand movements w.r.t head, face and torso.

3.2 Vision-based recognition system

Research on continuous sign language (CSL) started around mid-1990 off. Earlier studies did not account for the problem of MEs. As continuous sentences were not segmented into true signs and ME, – the, reported results were very poor [57, 83, 90, 91]. These early systems employed Hidden Markov Models (HMM), inspired by its success in speech recognition [40] and handwriting recognition [70].The next in line Conditional Random Field (CRF) models performed better than HMMs; but practical limitations and computational constraints limited the use of large scale CRFs. HMMs and CRFs solve the task of sign recognition from a probabilistic perspective. DP-inspired algorithms are capable of alignment and clustering of temporal sequences. Dynamic Time Warping (DTW) is a widely used DP-based algorithm, for gesture recognition that prevailed even before HMMs and CRFs. Recognition systems based on graphs are also adopted. Recent works in CSLR follow Deep Learning techniques. In the following sections, these methods are analyzed, in depth, in the context of SLR, and the systems that adopted these techniques are reviewed.

3.2.1 HMM-based CSLR

HMM is a statistical, generative model embedded in a Bayesian framework that depends on likelihood and priors; thus, the spatio-temporal variability in the input data stream can be effectively modeled with HMM [40]. For gesture sequence recognition, the trained HMMs corresponding to component signs are chained together to form a network of HMMs. The recognition problem is thus reduced to finding the optimal state sequence. Viterbi algorithm [18] finds the optimal state sequence, by implicit segmentation of the observation into its component signs, while computing a path through the chained HMM network. For more details on how to train an HMM and compute probability for output sequences, refer [78].

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Most of the earlier works on CSLR were HMM based, which offer the advantage of segmentation of a continuous gesture sequence into its constituent signs. The trickier problem of ME segmentation can be entirely avoided with HMM. Usually ME is modeled as an ergodic HMM (threshold model), where transitions are allowed from any state to any other state, and gestures are constructed as left-right HMM, where the transitions only go from one state to itself or to a unique follower. In this scenario, a sign is recognized only if the likelihood of best gesture model is greater than the threshold model.

The first work that dealt with a continuous sequence of gestures , by segmenting into gesture and non-gesture (ME) patterns, was by Lee and Kim [56]. The non-gesture patterns were handled by an artificial threshold model, which has an ergodic HMM architecture, having states copied from all gesture models (left-right HMMs). In this framework, a gesture is recognized only if likelihood of best gesture model is more than that of the threshold model. They carried out experimentation on a synthetic dataset having vocabulary size of 10 gestures; the results show a detection rate of 93.14%. The system poses a delay in predicting the label until the detection of next gesture. Volger et al. [93] solved the problem of modeling all possible combinations of phonemes by training Parallel HMMs (PaHMMs) independently, for separate channels, on a vocabulary of 22 signs. This is the earliest work in CSLR that explicitly modeled ME. For recognition of a sentence, Volger's team modified the token-passing algorithm to conform with PaHMM and merged the token probabilities, at phoneme and sign-levels, resulting in an equal recognition rate of 84.85%, at both levels. Similarly, Gao et al. [21] also explicitly modeled ME but their method was not vision-based. The HMM-based system by Holden et al. [39] achieved over 97% recognition rate on a sentence level, in a vocabulary of 21 words. But the same samples of 14 sentences were included in the training and testing sets. Testing on this kind of overlapping samples will not give an idea of the generalization capability of the model. In addition, they did not account for co-articulation or ME problems. Wenwen Yang et al. [112] employed HMM for computation of similarity between the sign model and candidate sign sequence, instead of DTW, since HMMs were much better in characterization of the sign. They embedded HMM into Level Building and enforced grammar and sign length constraints for CSLR. Besides, a fast algorithm for calculation of approximate HMM likelihood, called Fast-HMM, was proposed. Their FastHMM-based system, reported lower error rate of 12.2% in 9 seconds compared to [111], which resulted in error rate of 25.6% in 1.4 hours. Koller et al. [44] devised a large vocabulary CSLR system. For improved performance, the system incorporated signer-dependency by adaptation and language modeling of constrained maximum likelihood linear regression (CMLLR). The system handled ME as a non-gesture garbage HMM model. Variation in sign duration was tackled by length modeling and co-articulation by a universal transition model. System performance is evaluated on RWTH-PHOENIX [22] and SIGNUM [94] databases, and reported word error rate (WER) of 53% and 16.4% respectively. They have later adopted Deep Learning techniques for SLR. Zadghorban et al. [114] proposed an algorithm for Persian Sign Recognition, by first extracting the sign words from video, and then, classifying them based on motion and shape features of hands. They evaluated the performance on a synthetic Persian dataset, created as part of their work, and reported a recognition rate of 92.4

For recognition of signs from continuous data, an HMM is modeled per each sign label, with predefined number of hidden states s as shown in Fig. 2. Given a video clip as a sequence of images $X_1^N = X_1, X_2..X_n$, a CSLR system finds a sequence of sign words $Gloss_1^m$, for which X_1^N best fit the learned models. For decoding purpose, a sliding window approach is followed that considers each video segment from continuous utterance

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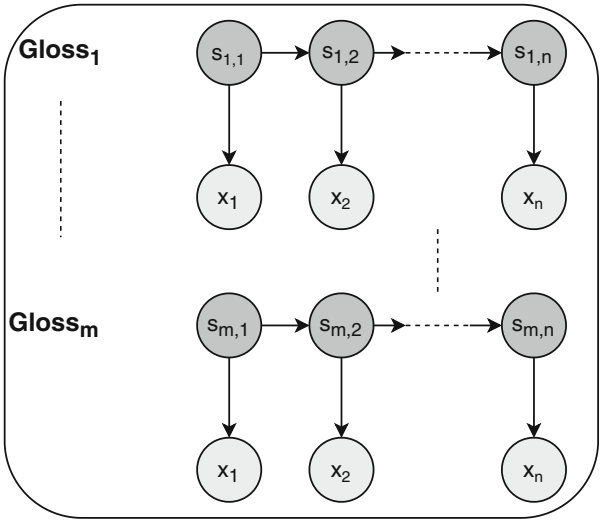


Fig. 2 HMM model for gesture/signs. For m labels and observation x of length n, x_j is an observation of input sequence x at position j and $s_{i,j}$ represents the state of HMM for sign label i ($Gloss_i$) and x_j

and Viterbi algorithm is carried out for each HMM, to find the most likely gesture, based on their scores. The model with best score will be considered as the sign (gloss or label). Mathematically, HMM tries to maximize the following equation:

$$[Gloss_1^m]_{optimal} = \arg\max_{Gloss_1^m} \left\{ \max_{s_i^n} \left\{ \prod_n p(X_n/s_n, Gloss_1^m) \cdot p(s_n/s_{n-1}, Gloss_1^m) \right\} \right\} \quad (1)$$

As the size of vocabulary increases, regular HMM fails to serve the purpose and models like TMMs have proved to be more effective [21].

As models of sequential data, HMMs have several limitations. This probabilistic model places more demands on the training data. Moreover, it can convey only $\log_2 N$ information about the recent history. The model is constrained by difficulties in learning long-range dependencies in a sequence, because of its low memory capacity [5]. In case of SLR, observations are not only dependent on the current state but also on the previous observations. As HMM assumes conditional independence between observations and strict independence across multivariate features, it does not allow representation of relationship between observations and labels across time. Researchers have proposed various extensions to HMMs for recognition of gestures/signs; such as factorial hidden Markov models (FHMMs) [29], coupled hidden Markov models (CHMMs) [8], parallel hidden Markov models (PaHMMs) [92] and Parametric HMMs (PHMMs) [102].

HMMs used for sign recognition follows first order Markov assumption that the next state is dependent only upon the current state. As this assumption is not applicable in case of CSLR, it may lead to degradation in overall performance. The option of modeling with n^{th} order HMMs that depend on previous n states, is not viable, due to the computational complexity of the model. Also, HMM works on Viterbi algorithm whose running time is $O(n^3)$. Considering all these facts, a better model that overcomes these drawbacks is one based on CRFs, which will be discussed in the coming section.

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3.2.2 DTW-based CSLR

DTW is a template-based dynamic-programming (DP) matching technique that has been applied to problems with temporal variability [60, 69]. It is useful for computation of similarity between two sequences of variable length and speed, limited to small vocabulary tasks. Suppose, the test sequence be represented as $i = (1, 2, .., I)$ and the reference sequence as $j = (1, 2, ..., J)$, on the assumption that both these sequences are segmented. Let $d(i, j)$ denote the local distance between a test frame and reference frame, and $D(i, j)$ be the accumulated distance from origin to current point (i, j). DTW aims at minimizing $D(i, j)$, that results in the best path (ordered gloss in case of CSLR). The accumulated distance is calculated as

$$D(i, j) = \Sigma_{k=1}^{K} W(i_k, j_k; i_{k-1}, jk - 1)d(i_k, j_k) \tag{2}$$

where (i_k, j_k) $(k = 1, ..K)$ represent a sequence of nodes and W is the transition cost between nodes. Comparing DTW and HMM, as their accuracies are context dependent, published literature have been ambivalent as to which of these is the preferred method, for a specific case [10].

Gao et al. [21] used the DTW method to match an input data with templates of ME, on the assumption that ME between two signs are always similar in different sentences. However, for real-world applications, ME varies between signs. For CSLR, Yang et al. [108] enhanced a classical Level Building algorithm, based on Dynamic Programming (DP), that enabled segmentation of sentences into signs and chunks of ME, without explicitly modeling the latter. Errors were controlled by enforcement of the trigram grammar model, which resulted in an accuracy of 83%. They evaluated hand motions only, ignoring hand shapes, for recognition of signs. This work was further enhanced by nesting DP within the modified Level Building algorithm to accommodate ambiguities in hand segmentation, emanated by changing context [111]. The enhanced model utilizes DTW for computation of distance between the sign model and test sign sequence, at each level, during the extensive search of matching gesture sequence. This time-intensive computation slows down the system. One advantage of such a matching process as adopted by [111] is the implicit segmentation of a sentence into signs foreclosing the burden of modeling ME, by classifying a segment as ME, when no good match could be found. For training the model base, only sign frames in continuous sentences are required, without the associated ME. Zadghorban et al. [114] proposed an algorithm for Persian Sign Recognition, by first extracting the sign words from video and then classifying them, based on motion and shape features of signer's hands. The authors evaluated the performance on a synthetic Persian dataset, created as part of their work and reported an accuracy of 92.3% with hybrid KNN-DTW classifier.

As the size of a vocabulary increases, more templates are required to map the variations in human gestures. In addition, they fail to handle undefined patterns. In such cases, HMM seems to be a more reliable option, as it can learn and adapt dynamic systems in real-time. Since DTW is a DP-based model, demand on training data is less and faster in static systems, where there is a fixed and specific standard for classification.

3.2.3 CRF-based CSLR

Conditional Random Fields (CRF), originally proposed in natural language processing [96], are discriminative models that directly compute the posterior probabilities [52, 54]. In CSLR, CRF is defined over a set of frames, where observations are the features $X = x_1, x_2.., x_n$ and the corresponding labels are $Gloss = gloss_1, gloss_2..., gloss_n$.

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$< Gloss, K >$ is a CRF, when X conditioned on $Gloss$, as modeled by a linear graph, obeys the Markov rule:

$$P(Gloss_i|X, Gloss - Gloss_i) = P(Gloss_i|X, N(Gloss_i)) \tag{3}$$

where $N(Gloss_i)$ are the neighbours of $Gloss_i$. In the linear graph G constructed by $< Gloss, K >$, let $C(Gloss, K)$ denote the set of cliques in G. Given the observation sequence X, the probability of a label sequence $Gloss$ according to the fundamental theorem of random fields is given by:

$$P(Gloss|X) \propto exp^{\Sigma_{c \in C(Gloss,X)} f_\theta F_\theta(c,X)} \tag{4}$$

where $\{F_\theta\}$ are the feature functions defined over all the cliques and $f = \{f_\theta\}$ are the parameters that act as weights assigned to edges in the graph. Unlike HMM, relationship between labels and observations can be represented, which allow inclusion of context-dependence and hence flexible. This ability of CRFs to represent relationship between labels and observations is a highly desirable feature for labeling any sequence with related context as in Continuous Sign Language. In contrast to HMM, Fig. 3 shows CRF, where a single model is constructed, for all sign labels.

Yang et al. [107] defined CRFs over a set of key frames, extracted from video of continuous sentences, and used them to segment sentences, in order to remove co-articulated frames. The investigators achieved 85% accuracy in segmentation of signs but did not lead to sign recognition. Yang et al. [109] developed a sign language spotting system with a threshold model based on a two-layer CRF (T-CRF). The system incorporated solutions for sub-sign reasoning, short-signs and appearance based sign verification. Solution to appearance based sign verification also aides to disambiguate signs with different hand shape but having similar overall hand motion. In a lexicon of 48 signs, the Yang system could detect signs from continuous data with a gesture spotting rate of 74.6%. In their next work [110], Yang's team developed a new approach for identification of signs and fingerspelling, from continuous utterance. A hierarchical framework of CRF and BoostMap embeddings are combined, to be categorized into signs and fingerspelligns, which can spot signs at the rate of 83%. These studies substantiated that CRF-based recognition systems could outperform HMM-based systems. CRFs are more powerful owing to the many parameters, that needs to optimized. So obviously CRFs require lot more training data. Practical limitations like demand for huge training data and computational constraints, limits its applicability.

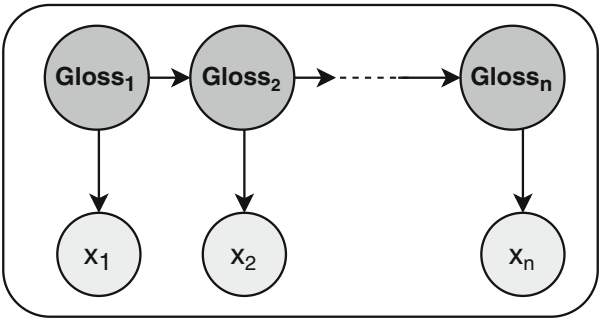


Fig. 3 CRF model for gesture/signs. For observation x of length n, $Gloss_j$ represents the label of input sequence x at position j

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Threshold model CRF has proved to be efficient to an extent for identifying ME [109, 110]. One of the variants of CRF is the hidden state conditional random field (HCRF) model [86]. HCRF has the combined ability of CRFs to use long range dependencies and of HMMs to model latent structure. In contrast to HCRF, Latent-Dynamic Conditional Random Field (LDCRF) model [68] could easily capture both sub-gesture patterns and dynamics between gesture, and also label unsegmented sequences.

3.2.4 Graph-based CSLR

Graphs are powerful to represent 3D data and model relationships. Yet, the complexity increases when it comes to temporal data and in such cases, the construction of graphs is an intricate ordeal. Nevertheless, researchers have come up smart workarounds to derive efficient utilization of this tool. The key advantage of graphs over other methods is offline extensibility of graph-model database, obviating the imperative of system re-training.

Anil Kumar et al. [3] used graphs for modeling 3D motion data. They created a 3D continuous dataset on a lexicon of 200 words. Each frame of a sentence in dataset is represented as a graph. Vertices of the graph are 57 spatial points of each frame and 56 joint pairs form the edges. Anil's team proposed an Adaptive Graph Match (GM) (AGM) algorithm, for verification of temporal matches between the query signs and sentences in a dataset. Intra-GM scheme was adopted to selectively extract motion frames, thereby discarding non-signs or ME frames; inter-AGM extracts signs in motion, leading to sign recognition by AGM. Their method obtained true word recognition frequency (TWRF) of 98.32%, on the synthetic dataset they have collected.

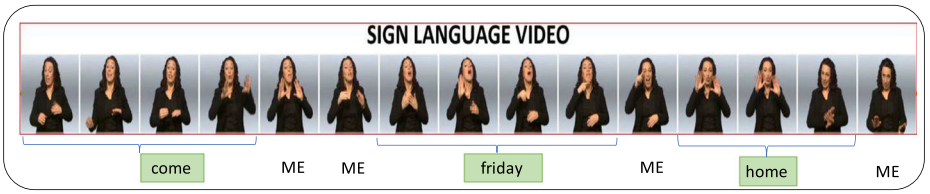
The model being computation-intensive, limits its application to real-time scenarios. Tolba et al. [88] built an Arabic SLR system wherein gestures corresponding to alphabets and words were represented as model graphs and a single decision tree connected all the graphs. They proposed a customized sub-graph iso-morphism algorithm for CSLR whose time complexity is bounded by a polynomial, irrespective of the size of graphs. On Arabic Sign Dataset, created as part of this work, they were able to achieve an accuracy of 80%.

All the four models - HMM, CRF, DTW, Graphical, which we have discussed till now, have adopted a common approach to deal with the problem of CSLR. Each model is trained on gestures of isolated signs (individual signs) and adapted to deal with the continuous signs. Also, ME has to be dealt with explicitly for correctly identifying continuous gesturing. Figure 4 gives an overview of how the tradition models work. Part (a) of the figure clearly shows that along with the glosses, ME segments are also recognized. Part (b) of the figure, shows a high level flow diagram of CSLR with the traditional techniques.

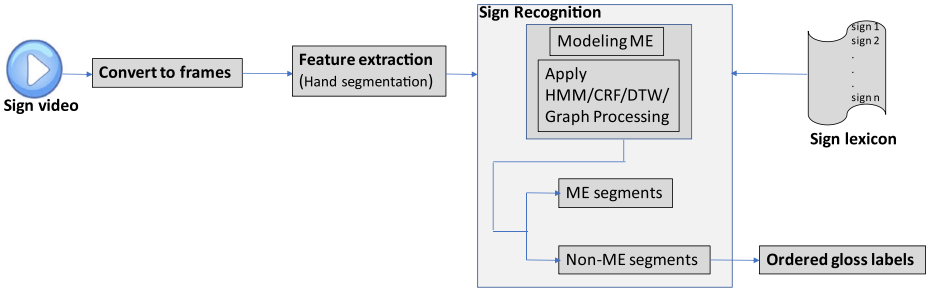
3.2.5 Deep learning-based CSLR

Deep learning (DL) is the prevailing state-of-the-art for any computer vision task, including SLR. In the context of SLR, Convolutional Neural Networks (CNN) [55] and variants of Recurrent Neural Networks (RNN) [76] - Long Short Term Memory Units (LSTM), are the widely used DL architectures [38]. The traditional models (HMM, CRF, DTW, Graph-based) tried to solve the task of CSLR by first dealing with ISLR (segmenting out ME from video) and then extended the learning to a complete sentence (sign video). But with DL, the problem is tackled directly, without applying any segmentation on the sign videos. The DL framework combines the visual representation learning and sequence modeling task, which takes as input, sign video with paired sentence-level annotation. While the visual representation learning is done by CNN, the sequence modeling task is carried out by LSTMs and

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(a) Traditional models identify the glosses as well as ME segments



(b) Flow diagram of continuous sign recognition with traditional techniques

Fig. 4 Traditional way of dealing with continuous signs

are jointly optimized. Various encoder-decoder models are designed with CNNs and LSTMs for sign recognition.

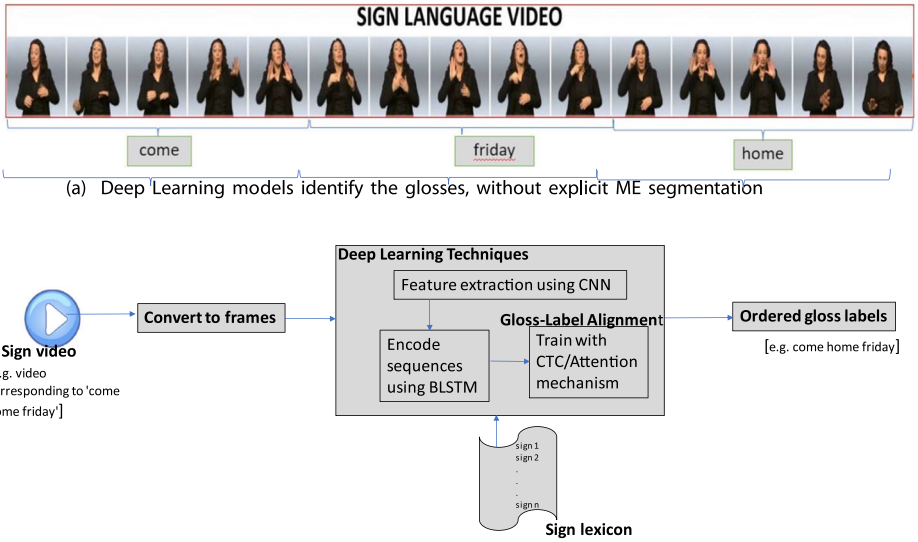
The emergence of CNN has removed the need for hand-crafted features and empowered recognition systems to implicitly learn the spatio-temporal features [11, 16]. CNN is mainly constituted of convolution (conv) and pooling layers. 2D conv-pooling is employed for learning 2D features whereas 3D conv-pooling is adopted for learning temporal features from videos. But CNN requires huge amount of labeled data for good generalization of features. Unsupervised or semi-supervised training could be resorted to in case of a shortage of labeled data. Moreover, techniques like transfer learning have proved to be beneficial in such cases. Regarding CSLR, CNN learns feature representations from the sign videos (full sentence) and final representations are taken from a convolution or pooling layer of our choice, from the CNN. Sign videos with paired sentence-level annotations are the inputs to CNNs.

The feature representations learned by a CNN are then fed to a sequence learning module. Variants of RNN are used for the task for sequence modeling. Vanilla RNNs are not capable of learning long-term dependencies and so in such scenarios, LSTM is utilized. With the help of a memory cell, LSTM can remember previous states, and hence, detect a sequence of patterns in time series. Although these properties enable LSTM to work in real-time, it does not consider the future data sequence for current prediction. This issue is easily tackled with Bidirectional LSTM (BLSTM) [11, 16]. Both past and future contexts are integrated into BLSTM during the bidirectional optimization process. BLSTM is trained with Connectionist Temporal Classification (CTC) [32] or attention mechanism for learning the word boundaries corresponding to each gloss in the video. The model tries to achieve correspondence between the input sequence and target transcription. For this purpose, first the features are extracted from sign videos and the sequence learning part is trained for

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alignment proposal. Then for better feature representations, the CNN is trained with the alignment proposals obtained. Thus, the end-to-end model is trained in an iterative fashion. Figure 5 gives an overall idea of what DL methodology is trying to achieve (part a) and how it achieves the goal (part b). RNNs have outperformed HMMs in sequence learning tasks [67], however with limited training data, RNNs are prone to overfitting which in turn fails to generalize the learning.

Koller et al. was the first group to adopt Deep Learning techniques for SLR. Their first work in this regard began by training a CNN on a million (10^6) hand images that were weakly labeled and used Expectation Maximization (EM) to integrate CNN with HMM for continuous sign recognition [45]. Their system was successfully generalized across datasets without retraining. They achieved a relative reduction in WER of over 17% on RWTH-PHOENIX-Weather and 24% on SIGNUM, when compared to [44]. Koller et al. extended this work to model cross-language sub-units using HamNoSys annotations, based on the modality of hand orientations [47]. The linguistically derived sub-units were aligned, to train the CNN model, and achieve additional improvement of the WER of a hand shape only system, by 2% absolute and a multimodal baseline by 0.5%. Koller et al. [46, 49] later modeled a hybrid system, called Deep Sign, of Convolutional Neural Network (CNN) embedded into HMM for CSLR. The initial frame state alignment for frame labeling was carried out as per the directives in [48], which generates an appropriate training-validation set for an optimal CNN model. CNN-HMM was trained in an end-to-end fashion, to accomplish a reduced WER on the public datasets, RWTH-PHOENIX and SIGNUM. The need for frame alignment was avoided with their latest "sequence to sequence" learning methodology using a series of specialized expert systems referred to as SubUNets [11]. The sequence to sequence model is formed of three tiers of neural networks - (1) CNN for feature extraction from images, (2) Bidirectional Long Short Term Memory Layers (BLSTM) for temporal modeling of the spatial features extracted by CNNs (3) CTC Loss Layer for training the network



(a) Deep Learning models identify the glosses, without explicit ME segmentation

(b) Flow diagram of continuous sign recognition with Deep Learning

Fig. 5 The DL way of dealing with continuous signs

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with videos of varying length and label sequences. The network is forced to model expert knowledge explicitly. The system could achieve results comparable to the state-of-the-art, without the need for frame alignment to segment signs for continuous recognition. Cui et al. [16] proposed a new model, comprised of CNN and BLSTM, the first fully deep learning based in this scope, which is trained in a three-stage optimization process for global sequence learning. For refining the results of sequence learning, a sliding window based detection network is also proposed. The system shows performance comparable to Deep Sign [46]. Differently to this work, later they adopted a different strategy for training the CNN + BLSTM network, [17]. At first the whole architecture was trained in an end-to-end fashion for alignment proposal, followed by fine-tuning of the feature extraction module. This was done in an iterative fashion to improve recognition results. This network takes on multi-modal data of RGB and optical flow images. This method when evaluated on the challenging sign datasets, SIGNUM and RWTH-PHOENIX-Weather 2014, achieved a reduction in WER of 4% and 2% respectively.

Very recently, Koller's team were finally able to go a step ahead by the implementation of sign language translation (SLT) [12]. SLT involves CSLR followed by translation to a spoken language, by including the grammatics. For this task, the authors modeled a framework of CNN-Encoder-Decoder-attention with an additional tokenization layer. Two ways of tokenisation were proposed - (1)Sign-To-Text (S2T): each frame was treated as a token and the translation was performed as an end-to-end system (2) Sign-to-Gloss-to-Text (S2G2T): each sign gloss was treated as a token; frame alignment similar to Re-Sign [48] was done at first to train the network, followed by gloss label prediction and used them for the final translation. The commonly used metrics for machine translations are BLEU [75] and ROGUE [63]. The S2T achieved 31.80 ROGUE, 32.24 BLEU-1, 19.03 BLEU-2, 12.83 BLUE-3 and the S2G2T achieved 43.80 ROGUE, 43.29 BLEU-1, 30.39 BLEU-2, 22.82 BLUE-3. The performance metrics are reported on a new dataset, RWTH- PHOENIX-Weather 2014T [12], which is an extension to RWTH- PHOENIX-Weather 2014 and meant solely for sign translation.

Even with the advancements in Deep Learning, the traditional methods are not fully obsolete for the task of sign recognition. In the works [45, 48] and [12], the DL models are combined with HMMs for improved frame alignments which in turn lead to overall performance spike of the recognition system. Even though CNNs are successful in feature learning and its ability to generalize, it is still an open problem as to how much training data (sign videos) is actually required to build a signer-independent system that can work in real-time. Till now all the works using DL have used 'videos of sentences' for training an end-to-end model. Unlike traditional methods of ME segmentation and continuous sign recognition, where ME segments were explicitly identified and eliminated, DL models have to implicitly do the word boundary identification as part of the training. This is done in the frame-label alignment step using optimization methods such as CTC or with attention weights in an Encoder-Decoder framework. Gloss to label mapping is then learned after doing frame alignments. So considering a complete sign recognition system, the performance completely relies on the gloss to label alignment part. CTC loss is now the widely used one for alignments between different languages. Since the continuous utterance varies in length, gloss to label alignment is very crucial for correctly identifying a sign sequence during testing. Complex backgrounds, co-articulation, short-signs, fingerspellings interleaved with dynamic signs, etc. are still challenging for DL models as well.

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4 Benchmark corpora

Benchmark datasets are available for the study of SLR, as standard reference, for researchers. It allows comparison of a model-free and signer-independent approaches [20]. Presently available benchmark databases, for evaluation of CSLR, are the following: Purdue RVL-SLLL [65], RWTH-PHOENIX-weather [22, 23], RWTH-BOSTON-104 and RWTH-BOSTON-400 [19], ATIS Sign Language Corpus [9], Corpus-NGT [15] and SIGNUM [94] (Table 1). Despite the availability of these standard corpora, only few researches in the area of SLR have referenced the databases. The results reported in most of the works reviewed are based on synthetic datasets. Among all the datasets, RWTH-PHOENIX-weather 2014 and SIGNUM are the widely used ones. The former features sign language recording of the daily news and weather forecast airings of the German public TV-station PHOENIX over three years. Since it is very much related to real-life data, sign recognition becomes more challenging. All recorded videos were at 25fps (frames per second) and the frame size is 210×260 pixels. SIGNUM dataset was created in a laboratory environment while carefully controlling the sign recording, making it less challenging compared to RWTH-PHOENIX-weather 2014. The videos were recorded at the rate of 30fps, with a frame resolution of 776x578.

5 Performance evaluation

In the literature of continuous sign language, performance of most of the recognition systems are measured using Word Error Rate (WER). Based on the levenshtein alignment between hypothesis and reference sentence, WER measures the required numbers of deletion, insertion and substitution operations to transform the recognized hypothesis into reference sequence. WER is calculated as

$$WER = \frac{\#insertions + \#deletions + \#substitutions}{\#reference\ observations} \tag{5}$$

Earlier works have reported results in terms of detection rate (R) given by,

$$R = \frac{\#correct\ spotting}{\#test\ signs} \tag{6}$$

Table 1 Statistics of publicly available benchmark databases that are commonly used for evaluation of Continuous Sign Language

Dataset	Year	Lexicon	#Signers	#Samples	Metric
Purdue RVL-SLLL	2002	-	14	2576	WER
RWTH-PHOENIX-weather	2012	911	7	1980	WER
RWTH-PHOENIX-weather	2014	1558	7	6861	WER
RWTH-BOSTON-104	2008	168	3	201	WER
RWTH-BOSTON-400	2008	406	4	843	WER
ATIS	2008	400(ISL)	6	595	WER
Corpus-NGT	2008	3900	92	116	WER
SIGNUM	2007	450	25	14000	WER

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Some of the recent works have used Word Recognition Frequency (WRF) and Word Matching Score (WMS), expressed as below.

$$WRF = \prod_{i=1}^{g} \frac{N(S_s^i)}{N(S_g)} \tag{7}$$

where $N(S_s^i)$ is number of times a sign (S) in the test video matches with those in dataset groups g (formed of multiple start and end frames for each word) and $N(S_g)$ is the number of signs in the groups. This metric is specific for Adaptive Graph Matching technique proposed by Anil Kumar et al. [3], where graphs corresponding to frames are grouped for matching purpose.

$$WMS = \frac{\#Correct\ classifications}{Total\ signs\ inputted\ to\ classifier} \tag{8}$$

These measures for all the reviewed works are specified in the Tables 2 and 3.

The DL models for CSLR with their performance on benchmark datasets are tabulated in Table 2. Table 3 presents an overview of the traditional methods that have employed hand-crafted features. From Table 3, it is clear that only Koller et al. [44] have used benchmark dataset for performance evaluation. All other works have used synthetic dataset and the average detection rate or accuracy reported is between 80-90%. Just the reverse considering the other table (Table 2) i.e. almost all works report performance on standard dataset and the average detection rate (i.e.$100 - WER$) is only between 60-70%. The huge difference in detection rate is basically due to the datasets, that the accuracy was reported on. The synthetic datasets that the traditional models reported accuracy were less challenging in terms of background, signer-dependent variations, overlapping signers between train and test, clear pauses between signs which leads to easier ME segmentation etc.. But the benchmark datasets which the DL models reported acccuracies are more related to real life data and therefore challenging and hence lower detection rate compared to traditional techniques. Comparing the size of training data used by traditional and DL models, the former mostly worked on small synthetic datasets and the latter were trained on huge datasets. Correspondingly, a performance difference between the two approaches is noticeable. As the amount of training data increases beyond a certain number, continuous sign recognition rate of traditional approaches becomes steady, whereas for DL approaches, performance increases with respect to the increment in the amount of data, as shown in Fig..6.

Another noticeable performance variation in Table 3 is with the HMMs. The results reported on synthetic datasets with HMMs are very appealing but when evaluated on benchmark datasets like RWTH-PHOENIX and SIGNUM [44], there is a sharp drop in the results. This clearly shows that models like HMM fails to be applicable for more real life data. CRF is a better model than HMM but due to its complexity in handling larger utterances, no works have reported results of CRF on benchmark datasets. Similar limitations of DTW and graph based techniques is the primary reason behind not using them for evaluation on standard corpora.

For all the DL and traditional methods, time complexity is computed and recorded in the Tables 2 and 3. HMM, CRF and DTW based works report the highest complexity which is of order n^3 but with Fast-HMM [112], the complexity has been reduced to order of n^2. For DL, the time complexity is found to be $O((\sum_{l=1}^{d}(n_{l-1}.s_l^2.n_l.m_l^2) + w))$, where d is the

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Table 2 Deep Learning solutions for CSLR

Method	Deep model	Description	Dataset	Score	Complexity
Koller et al. [45]	CNN[1]+HMM[2]	Trained on 1million hand shapes	RWTH-PHOENIX-Weather 2014 SIGNUM	45.1% 7.6% (WER[7])	$O(n^3)$
Koller et al. [47]	CNN+HMM	Hand shape only system Multi modal b aseline	RWTH-PHOENIX-Weather 2014 RWTH-PHOENIX-Weather 2014	48.2% 45.5% (WER)	$O(n^3)$
Koller et al. [46]	Hybrid CNN+HMM	Initial frame-state alignment done	RWTH-PHOENIX-Weather 2014 RWTH-PHOENIX-Weather 2012 SIGNUM	38.8% 30% 7.4% (WER)	$O(n^3)$
Cui et al. [16]	CNN+BLSTM[3]+CTC[4]	No explicit alignment done	RWTH-PHOENIX-Weather 2014	38.7% (WER)	$O((\sum_{l=1}^{d}(n_{l-1}.s_l^2.n_l.m_l^2)+w).i.e)$ d-#convolutional layers n_l-#filters in l^{th} layer n_{l-1}-#input channels of lth layer s_l-spatial size of filter m_l-spatial size of output w-#weights of LSTM i-input length e-#epochs

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Table 2 (continued)

Method	Deep model	Description	Dataset	Score	Complexity
Cui et al. [17]	CNN+BLSTM+CTC	Iterative training with segment-gloss alignment	RWTH-PHOENIX-Weather 2014 SIGNUM	22.86% 2.8% (WER)	O($(\sum_{l=1}^{d}(n_{l-1}.s_l^2.n_l.m_l^2)+w).i.e$) d-#convolutional layers n_l-#filters in l^{th} layer n_{l-1}-#input channels of lth layer s_l-spatial size of filter m_l-spatial size of output w-#weights of LSTM i-input length e-#epochs
Koller et al. [48]	CNN+2BLSTM+HMM	Re-alignments are done iteratively	RWTH-PHOENIX-Weather 2014 SIGNUM	26.8% 4.8% (WER)	O($(\sum_{l=1}^{d}(n_{l-1}.s_l^2.n_l.m_l^2)+w).i.e$) d-#convolutional layers n_l-#filters in l^{th} layer n_{l-1}-#input channels of lth layer s_l-spatial size of filter m_l-spatial size of output w-#weights of LSTM i-input length e-#epochs

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Table 2 (continued)

Method	Deep model	Description	Dataset	Score	Complexity
Koller et al. [11]	CNN+BLSTM+CTC	Alignment & recognition is jointly learned from hand shape sequences	RWTH-PHOENIX-Weather 2014	40.7% (WER)	O($(\sum_{l=1}^{d}(n_{l-1}.s_l^2.n_l.m_l^2)+w).i.e$) d-#convolutional layers n_l-#filters in l^{th} layer n_{l-1}-#input channels of lth layer s_l-spatial size of filter m_l-spatial size of output w-#weights of LSTM i-input length e-#epochs
Ananth Rao et al. [2]	ANN[5]	ME not handled	(Synthetic) Vocab=20 #Videos=180	90% (WMS[8])	O($e.m.f.n$) e-#epochs m-#examples f-#features n-#neurons
Koller et al. [49]	Ensembles of Hybrid CNN+HMM	Classification relies purely on the right hand	RWTH-PHOENIX-Weather 2014 RWTH-PHOENIX-Weather 2012 SIGNUM	32.5% 30% 7.4% (WER)	O(n^3)

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Table 2 (continued)

Method	Deep model	Description	Dataset	Score	Complexity
Jie Huang et al. [41]	LS-HAN[6]	Frame clips to word alignment done by HAN	RWTH-PHOENIX-Weather 2014	38.4% (WER)	O(($\sum_{l=1}^{d}$ ($n_{l-1}.s_l^2.n_l.m_l^2$)+ w).i.e) d-#convolutional layers n_l-#filters in l^{th} layer n_{l-1}-#input channels of lth layer s_l-spatial size of filter m_l-spatial size of output w-#weights of LSTM i-input length e-#epochs0

[1] Convolutional Neural Network

[2] Hidden Markov Model

[3] Bidirectional Long Short Term Memory

[4] Connectionist Temporal Classification

[5] Artificial Neural Network

[6] Hierarchical Attention Network with Latent Space

[7] Word Error Rate

[8] Word Matching Score

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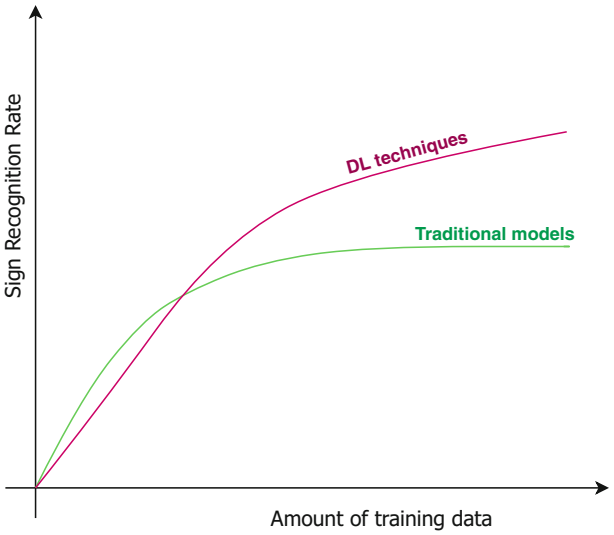


Fig. 6 Performance variation of traditional methods vs Deep Learning techniques, with increasing training data

number of convolutional layers, n_l is the number of filters in l^{th} layer, n_{l-1} is the number of input channels of lth layer, s_l is the spatial size of filter, m_l is the spatial size of output, w is the weights of LSTM, i is the input length and e is the number of epochs. Compared to n_{l-1} and m_l, other parameters can be assumed to be of constant size and the resulting complexity will be of the order of n^2.

Figures 7 and 8 shows the performance of different works on the challenging RWTH-PHOENIX-weather 2014 dataset and SIGNUM. Performance of recognition systems, on the datasets, is found to improve over time, with DL techniques. Various encoder-decoder models of CNN and LSTM, with varying training methodologies, led to the performance spike on the challenging datasets. The lowest WER reported on SIGNUM is 2.8 and on

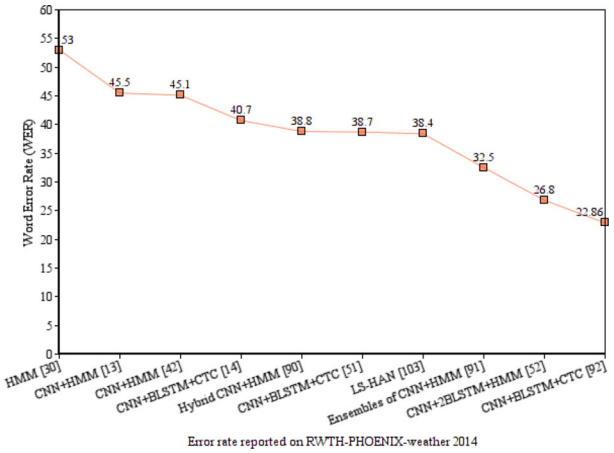


Fig. 7 Error rate reported on RWTH-PHOENIX-weather 2014

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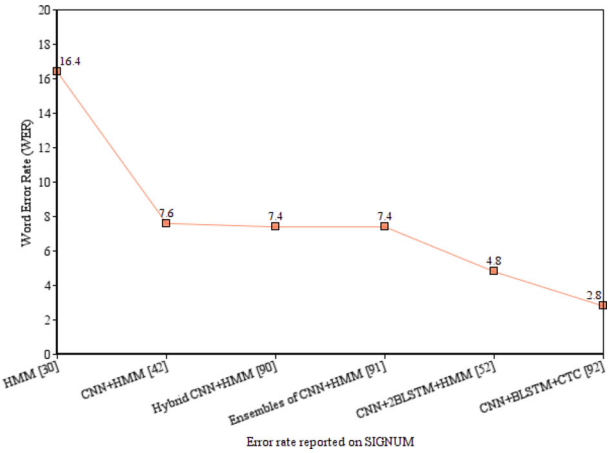


Fig. 8 Error rate reported on SIGNUM

PHOENIX 2014 is 22.86. This is because the word distribution count in both datasets are different i.e. PHOENIX has a large number of words with single occurrence whereas in SIGNUM the least frequent word count is 3. This data distribution of SIGNUM favours training and shows its artificial characteristics which leads to very low WER.

6 Discussion

Feature extraction is explicitly hand-engineered for HMM/CRF/DTW-based models. This tedious task of feature extraction is avoided with Deep Learning methods. Most of the traditional works are HMM-based, even though CRF was proved to be more efficient. Both have their limitations when scaled to large vocabularies. HMMs follow the first order Markov assumption, that makes it incompetent to store long range dependency information, which is an essential requirement for continuous sign recognition. The complexity of CRFs when applied to long utterances and its demand for huge training data has limited is application for the task of sign recognition. Similar complexity is associated with graphs and DTW, when it comes to dealing with temporal data. The state-of-the-art DL methodology makes use of CNN for inherent feature extraction from videos and LSTM for learning the sequential and temporal dynamics. Comparing Tables 2 and 3, it is evident that the traditional works were limited to synthetic databases, created in a very constrained manner to ease the process of feature extraction. With the advent of DL, it is clear from Table 2 that the latest works are evaluated on benchmark datasets which are very close to real-life data. This is due to the elimination of the main difficulty of feature engineering, by DL. The required features are learned implicitly at the cost of huge training data. Nowadays even with small datasets, accurate feature representation is possible with the technique of transfer learning. The advancements in hardware like GPU also adds to the popularity of DL. Thus more researchers were naturally attracted to vision-based methods of dealing with CSLR than Dataglove-based methods since the main difficulty of feature engineering that prompted researchers to go for the latter has been eliminated. This fact is proved in Fig. 9, which shows the graphical representation of the number of vision-based works that dealt with CSLR per

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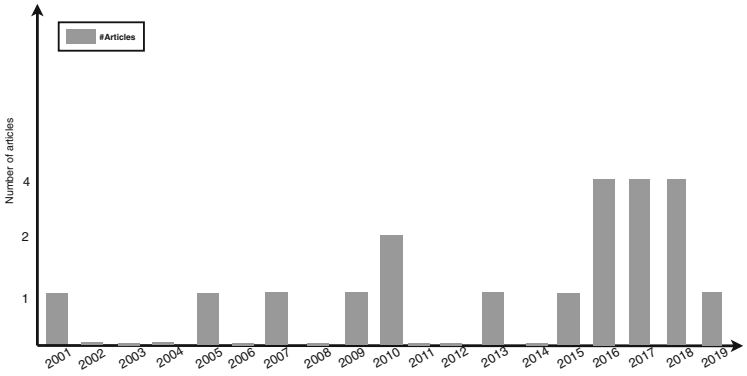


Fig. 9 Number of works related to vision-based CSLR per year

year has gone up owing to the breakthroughs in DL. Figure 10 shows the taxonomy of this whole study of vision-based approaches for CSLR.

To correctly decode a continuous sign including its grammar, both manual and non-manual signs need to be integrated. This aspect has not been dealt with efficiently, even with the DL methodology. From our survey, it is clear that inadequate importance has been given to the grammatical aspect of sign language. To deal with more real life data, complex background processing is a major hurdle. A possible solution is to use the concept of saliency detection [25, 115] to get the salient portions from the video frames. This comes as a pre-processing step before the feature extraction process. An additional concern is scaling up of sign vocabularies. The dataset field of Tables 2 and 3 shows that the largest vocabulary has 1558 sign glosses (of RWTH-PHOENIX-weather 2014). The size is small compared to related tasks in Natural Language Processing. The feasibility of building a commercial CSLR system of a large lexicon, to work in real-time, is still an open area of research.

Another possible research direction is the recognition of signs in real-time from continuous streaming data. Working with real-time data involves slightly different mechanisms as compared to working with historical data. No research has yet been reported in streaming analytics, in the field of Sign Language. The state-of-the-art DL models cannot be applied directly for this task, since BLSTMs require full sentence to learn the context. Also, LSTMs are very slow in processing and consumes much of the resources, which makes it practically unsuitable for real-time processing and ultimately limits the applicability of neural networks

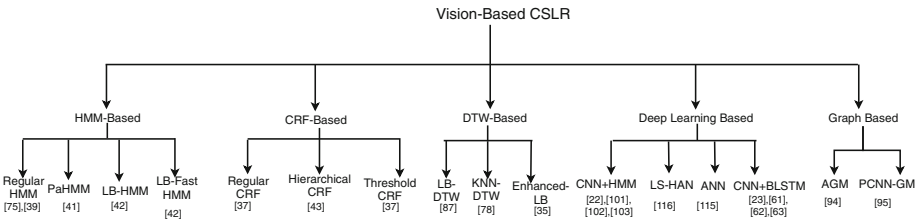


Fig. 10 Taxonomy tree of vision-based methods for CSLR. PaHMM- Parallel HMM, LB- Level Building, CNN- Convolutional Neural Networks, BLSTM - Bidirectional LSTM, AGM - Adaptive Graph Matching, LS-HAN - Hierarchical Attention Network with Latent Space, PCNN - Pulse Coded Neural Network, GM - Graph Matching

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Table 3 Traditional hand-crafted features based solutions for CSLR

Method	Model	Description	Dataset	Measure	Score	Complexity
Volger et al. [93]	PaHMM[1]	Explicitly modeled ME	(Synthetic) Vocab=22 #Videos=499	Detection rate	84.5%	O(n^3)
Holden et al. [39]	HMM[2]	Do not deal with co-articulation or ME problems	(Synthetic) Vocab=21 #Videos=415	Detection rate	97%	O(n^3)
Ruiduo Yang et al. [108]	Dynamic Programming based Level Building	No explicit model for ME	(Synthetic) Vocab=39 #Videos=25	WER[3]	75%	O(n^3)
Ruiduo Yang et al. [111]	Enhanced Level Building based on Nested Dynamic Programming	No explicit model for ME	Subset of Purdue	Detection rate	80%	O(n^3)
Wenwen Yang et al. [112]	Level Building based on Fast HMM	Threshold-HMM for identifying ME	(Synthetic) Chinese SL Vocab=21 #Videos=100	WER	12.2%	O(n^2)
Hee-Deok Yang et al. [109]	2layer CRF[4] - An adaptive Threshold (T)-CRF + CRF	Adaptive T-CRF to distinguish signs & non-signs (ME)	(Synthetic) Vocab=48 #Videos=98	Detection rate	74.6%	O($mNTQ^2nS$) m-#training iterations N-#training data sequences T-average length of training sequences Q-#class labels n-#CRF features S-search time of the optimization algorithm

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Table 3 (continued)

Method	Model	Description	Dataset	Measure	Score	Complexity
Hee-Deok Yang et al. [110]	Hierarchical CRF + BoostMap embeddings	Adaptive T-CRF to distinguish signs & non-signs (ME)	(Synthetic) Vocab=24 #Videos=98	Detection rate	83%	O($mNTQ^2nS$) m-#training iterations N-#training data sequences T-average length of training sequences Q-#class labels n-#CRF features S-search time of the optimization algorithm
Koller et al. [44]	HMM	Universal Transition Model to account for ME	RWTH-PHOENIX SIGNUM	WER WER	53% 16.4%	O(n^3)
Anil Kumar et al. [3]	AGM[5]	Intra-GM to extract signs alone, discarding ME	(Synthetic) Vocab=200 #Videos=121	TWRF[6]	98.32%	-
Zadghorban et al. [114]	Hybrid KNN-DTW	SL video decomposed to sign words	(Synthetic) Vocab=46 #Videos=300	Detection rate	92.3%	O(n^3)

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Table 3 (continued)

Method	Model	Description	Dataset	Measure	Score	Complexity
Tolba et al. [88]	Hybrid PCNN[7] and GM[8]	ME removed by filtering hand postures after hand classifications	(Synthetic) Vocab=50 #Videos=30	Detection rate	80%	-

[1] Parallel Hidden Markov Model

[2] Hidden Markov Model

[3] Word Error Rate

[4] Conditional Random Fields

[5] Adaptive Graph Matching

[6] Word Recognition Frequency

[7] Pulse-Coupled Neural Network

[8] Graph Matching

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solutions. The best alternative is to employ a model that require less parameters to reduce memory access or increase re-usability for multiple computations.

7 Conclusion

Major recent works, in the area of vision-based continuous sign recognition, were covered in this survey paper. HMM was the baseline model for most of the works. Few studies used CRF and DTW. Deep Learning is the state-of-the-art for CSLR. Challenges and benchmark datasets for the task of recognition of signs, were also detailed in this work. While the main challenge of ME was dealt with explicitly in case of HMM/CRF/DTW, the problem is solved implicitly with the frame-label alignment technique of DL. The problems of co-articulation, short-signs, complex backgrounds and fingerspellings interleaved with dynamic signs are still an open issue, even with DL. The existing sequence learning DL model, LSTM, is slow and computation-intensive, that limits its application to large vocabulary, real-time systems. Also, for real-time recognition of signs from continuous streaming data, the existing DL models will fail to meet the requirements. It is worthy of note that not much importance has been given to non-manual features, which play a major role in the grammatical aspects of continuous signs. Increasing the size of lexicon is another open challenge. Hence, several potential research directions could be easily identified from our survey, indicating numerous opportunities in this field, despite the advances achieved to date.

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